

jQuery Complete Course (Basic to Advanced)

Module 1: Introduction to jQuery

Lesson 1

- What is jQuery?
- History of jQuery
- Why jQuery?
- Advantages of jQuery
- Features of jQuery
- jQuery vs JavaScript
- CDN vs Local Installation

Add jQuery using CDN

```
<script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>
```

Lesson 2

First jQuery Program

```
$(document).ready(function(){  
    alert("Welcome to jQuery");  
});
```

Short Syntax

```
$(function(){  
  
});
```

Topics

- Document Ready
- Why document ready is important

Module 2: jQuery Selectors

Lesson 3

Basic Selectors

- Element Selector
- ID Selector
- Class Selector
- Universal Selector
- Multiple Selectors

Examples

`$("p")`

`$("#heading")`

`$(".container")`

`$("*")`

`$("h1,p")`

Lesson 4

Advanced Selectors

- first
- last
- even
- odd
- eq()
- gt()
- lt()
- contains()
- empty
- has()

Projects

Highlight Table Rows

Module 3: Events

Lesson 5

Mouse Events

- `click()`
- `dblclick()`
- `mouseenter()`
- `mouseleave()`
- `hover()`

Projects

Image Hover Effect

Lesson 6

Keyboard Events

- `keypress()`
- `keydown()`
- `keyup()`

Projects

Character Counter

Lesson 7

Form Events

- `submit()`
- `change()`
- `focus()`
- `blur()`

Projects

Registration Form Validation

Module 4: Effects

Lesson 8

Show & Hide

show()

hide()

toggle()

Projects

FAQ Accordion

Lesson 9

Fade Effects

- fadeIn()
- fadeOut()
- fadeToggle()
- fadeTo()

Projects

Image Gallery

Lesson 10

Slide Effects

- slideDown()
- slideUp()
- slideToggle()

Projects

Dropdown Menu

Module 5: Animations

Lesson 11

animate()

```
$("#box").animate({  
  left:"300px"  
});
```

Queue

stop()

delay()

Projects

Moving Box

Module 6: DOM Manipulation

Lesson 12

Methods

- html()
- text()
- val()
- attr()
- prop()

Projects

Student Registration Form

Lesson 13

CSS Manipulation

Methods

- `css()`
- `addClass()`
- `removeClass()`
- `toggleClass()`
- `hasClass()`

Projects

Dark Mode Toggle

Lesson 14

DOM Manipulation

- `append()`
- `prepend()`
- `before()`
- `after()`
- `empty()`
- `remove()`

Projects

Dynamic Todo List

Module 7: Traversing

Lesson 15

Methods

- `parent()`
- `children()`
- `siblings()`
- `next()`
- `prev()`

Projects

Lesson 16

More Traversing

- `find()`
- `filter()`
- `not()`
- `closest()`

Projects

Search Highlight

Module 8: Dimensions

Lesson 17

Methods

- `width()`
- `height()`
- `innerWidth()`
- `outerWidth()`
- `offset()`
- `position()`

Projects

Responsive Layout Checker

Module 9: AJAX

Lesson 18

Introduction

- What is AJAX?
- AJAX Workflow

Methods

- load()
- get()
- post()

Projects

Load Student Data

Lesson 19

JSON with AJAX

Methods

\$.getJSON()

Projects

Employee Information

Lesson 20

API Integration

Using Public APIs

Projects

- Weather App
 - Random User Generator
-

Module 10: Forms

Lesson 21

Form Validation

- Required Fields
- Email Validation
- Password Validation

Projects

Registration Form

Module 11: Chaining

Lesson 22

Method Chaining

```
$("#box")  
.hide()  
.fadeIn()  
.slideDown();
```

Projects

Animated Card

Module 12: Callbacks

Lesson 23

Callback Functions

Example

```
hide(1000,function(){  
  
});
```

Projects

Popup Animation

Module 13: Plugins

Lesson 24

Popular Plugins

- Owl Carousel
- Slick Slider
- DataTables
- Lightbox
- Select2

Projects

Image Slider

Module 14: Deferred Objects

Lesson 25

- Promise
- Deferred
- when()

Projects

Multiple AJAX Requests

Module 15: Utility Functions

Lesson 26

Methods

- each()
- grep()

- map()
- trim()
- type()

Projects

Student Marks Processing

Module 16: Performance

Lesson 27

Optimization

- Event Delegation
- Caching Selectors
- Efficient DOM Updates

Best Practices

Module 17: Mini Projects

Lesson 28

Projects

- Digital Clock
 - Image Gallery
 - FAQ Section
 - Todo List
 - Login Form
-

Module 18: Advanced Projects

Lesson 29

Projects

- Weather App
 - Student Management System
 - Employee CRUD
 - Shopping Cart
 - Quiz Application
-

Module 19: Final Project

Lesson 30

Build a Complete Interactive Website

Pages

- Home
- About
- Services
- Portfolio
- Gallery
- Contact

Features

- Smooth Scrolling
 - Sticky Navbar
 - Image Slider
 - Contact Form Validation
 - AJAX Contact Form
 - Dark Mode
 - Animations
 - Loading Spinner
 - Scroll to Top Button
-

30 Hands-on Practice Projects

1. Welcome Alert
2. Hide & Show Text
3. Toggle Button
4. Change Background Color

5. Image Hover Effect
 6. Character Counter
 7. Registration Form
 8. Login Form Validation
 9. FAQ Accordion
 10. Animated Box
 11. Image Slider
 12. Student Registration Form
 13. Todo List
 14. Shopping Cart UI
 15. Search Filter
 16. Employee Table
 17. Dynamic Table
 18. Weather App
 19. Random Quote Generator
 20. Digital Clock
 21. Stopwatch
 22. Countdown Timer
 23. Expense Tracker
 24. Quiz App
 25. Notes Application
 26. Photo Gallery
 27. Restaurant Menu Filter
 28. Portfolio Website
 29. Admin Dashboard
 30. Interactive Company Website
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jQuery Interview Questions (Basic to Advanced)

Beginner

1. What is jQuery?
2. What are the advantages of jQuery?
3. Difference between JavaScript and jQuery?
4. What is the `$()` function?
5. What is `$(document).ready()`?
6. What are selectors in jQuery?
7. Explain event handling.
8. Difference between `html()` and `text()`?
9. Difference between `attr()` and `prop()`?
10. Explain method chaining.

Intermediate

11. What is AJAX in jQuery?
12. Difference between `$.get()` and `$.post()`?
13. Explain event delegation.
14. What are callback functions?
15. What is `each()`?
16. Explain `append()` vs `prepend()`.
17. What are jQuery plugins?
18. How do you use `animate()`?
19. Difference between `fadeIn()` and `slideDown()`?
20. What is `closest()`?

Advanced

21. Explain Deferred Objects.
 22. What is `$.when()`?
 23. How do you optimize jQuery performance?
 24. Explain event bubbling with jQuery.
 25. What is DOM traversal?
 26. Difference between `offset()` and `position()`?
 27. Explain AJAX error handling.
 28. How do you create a custom jQuery plugin?
 29. What are the limitations of jQuery in modern web development?
 30. When should you use vanilla JavaScript instead of jQuery?
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15 Real-World Mini Projects

1. Responsive Navigation Bar
2. Login Form Validation
3. Registration Form
4. Student Management System
5. Employee CRUD UI
6. Weather Application
7. Random User Generator
8. Product Filter
9. Image Gallery
10. FAQ Accordion
11. Shopping Cart UI
12. Portfolio Website
13. Admin Dashboard
14. Restaurant Menu Filter
15. Interactive Company Website

jQuery Cheat Sheet

Include jQuery

```
<script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>
```

Selectors

```
$("#p")  
$("#id")  
$(".class")  
$("*")  
$("h1,p")
```

Events

```
.click()  
.dblclick()  
.hover()  
.mouseenter()  
.mouseleave()  
.keyup()  
.submit()  
.change()
```

Effects

```
.show()  
.hide()  
.toggle()  
.fadeIn()  
.fadeOut()  
.slideDown()  
.slideUp()  
.animate()
```

DOM Manipulation

```
.html()  
.text()  
.val()  
.attr()  
.prop()  
.css()  
.addClass()  
.removeClass()  
.toggleClass()
```

.append()
.prepend()
.remove()
.empty()

AJAX

\$.get()
\$.post()
\$.ajax()
\$.getJSON()
.load()